United States of America
Before The
Federal Energy Regulatory Commission

Interconnection of Large Loads to the Interstate Transmission System	} } } }	**Docket No. RM26-4-000**

Comments of Equinix, Inc.

I. Introduction

Equinix, Inc. ("Equinix") is pleased to have this opportunity to offer its preliminary comments on the important topics at issue in this docket. We anticipate this proceeding will raise urgent and critical questions central not only to our nation's energy system and the interaction of federal, state and local jurisdictions over it, but to our overall economy, quality of life and global competitiveness.

II. Background

A. Procedural Background

On October 23, 2025, the Secretary of Energy issued a letter including a proposal for an Advance Notice of Proposed Rulemaking (the "Proposed ANOPR"), pursuant to Section 403 of the Department of Energy Organization Act, 42 U.S.C. § 7173(a). On October 27, 2025, the Federal Energy Regulatory Commission ("Commission" or "FERC") issued a Notice Inviting Comments in this docket (the "Notice"). On November 7, 2025, the Commission issued a Notice Granting Extension of Time, providing until November 21, 2025, for submission of initial comments on the proposed ANOPR (the "Extension of Time Notice"). Equinix hereby submits these initial comments pursuant to the Notice and the Extension of Time Notice.

B. Background on Equinix

Equinix, headquartered in Redwood City, California, is the world's digital infrastructure company. We operate over 270 data centers across six continents, 37 countries and 76 metropolitan areas around the globe. We provide roughly 500,000 data interconnections to over 10,000 customers, including the world's largest cloud platforms, financial institutions, artificial intelligence ("AI") developers, and public sector bodies. Equinix is the global leader in data center colocation services, which grows increasingly central to the way the modern economy works by ensuring secure, low-latency, and highly resilient data exchange. Equinix is excited and proud to deliver infrastructure that advances the nation's AI leadership, economic development, and strategic competitiveness.

III. Communications

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IV. Comments

The Proposed ANOPR is founded on the importance to the nation of AI and the broader digital economy that flow through data centers, as well as other large loads such as the return of domestic manufacturing.[1] If data centers in this country face severe headwinds to energization, especially relative to competing nations, it is no exaggeration to say growth and innovation in all sectors of the economy may be hobbled, and our global competitiveness diminished. We applaud the Department of Energy and the Commission for taking the initiative on these issues, and for commencing this proceeding.

At this juncture, we are pleased to offer some background for the Commission's consideration on the considerable diversity of data center types and business models, as well as a suggested organizing principle and initial threshold questions. We hope these contributions are helpful to the Commission in identifying priorities and developing non-discriminatory solutions that promote a more reliable, adequate and affordable energy system while also supporting the national priorities that the Secretary of Energy identified in his letter to the Commission.[2] Equinix otherwise reserves further comment at this time and looks forward to participating as this proceeding progresses.

A. Non-Discriminatory Large Load Interconnection Rules Should Reflect the Diversity of Large Loads and of Their Interaction with the Energy System

As part of its commitment to the nation's leading position in AI and the digital economy, Equinix is rapidly scaling its data center capacity and will be bringing multiple new data centers online, including some of the largest in our history, in the coming years. As such, Equinix has a clear interest in the timely and adequate energy interconnection[3] of those projects to the grid.

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[1] Letter from Secretary of Energy Chris Wright at 2 (Oct. 23, 2025), *available at https://www.energy.gov/sites/default/files/2025-10/403%20Large%20Loads%20Letter.pdf.*
[2] *Id.*
[3] One note on terminology that may be helpful as this proceeding progresses. Many terms, including "colocation" and "interconnection," are used extensively in the data and energy sectors, but have very different meanings for each sector. The terms "energy interconnection" and "energy colocation" and "data interconnection" and "data colocation" may be helpful to differentiate these terms, although they are not in common usage in the energy or data sectors.

Equinix' interest is not limited to its own behalf, but on behalf of the clients it serves in those data centers, which, as noted above, range from including the world's largest cloud platforms, financial institutions, AI developers, communications companies, entertainment companies, and public sector bodies.

The range of data center clients, as broad as they are, does not begin to explain how important data centers have become to our economy, way of life and national security. Virtually every financial transaction other than cash; key aspects of our education, medical, and other social services; the communications and entertainment we enjoy; and, of greatest relevance to the Commission's domain, a significant and growing proportion of the regulation and operations of the grid itself flow through data centers (including, we might add, comments filed in this docket). Data centers are increasingly the medium through which our economy, including much of the energy sector, and much of our daily lives occurs.

Data centers are extremely diverse, in terms of physical plant configurations, business models, geographical location, and relative proximity to load centers. Much of the recent press on data centers has focused on one type, the "hyperscale" dedicated AI training facilities that are increasingly planned in terms of gigawatts of energy need rather than megawatts. As those types of data centers are designed to develop AI models, rather than enabling provision of services to customers with demanding needs, they have greater flexibility in terms of location and may be located relatively far from energy load centers where land and often energy may be more accessible. However, the vast majority of data centers, such as colocation and edge data centers, must be located as close to the businesses and public entities that they serve as possible. Unsurprisingly, those locations are generally in the heart of load centers.

It may serve the Commission's interests and the issues central to this docket to more deeply explore the broad and diverse taxonomy of data centers and their business models as this proceeding progresses. Since colocation data centers are so central to the economy but are much less well-known outside of the data sector itself, we offer a brief explanation.

Colocation facilities allow a diverse range of entities, from smaller firms and public entities to Fortune 500 companies, to house *their own* IT infrastructure within a single energy-efficient, secure, professionally and neutrally managed environment. The most densely connected colocation data centers house hundreds of customers and thousands of data interconnections. Physical proximity of the colocation facilities to customer locations, and of the customers' own IT infrastructure to each other, allows the swift and secure interchange of data essential to the operation of modern economies. Within data colocation facilities, customers can "peer" or directly interconnect enterprise-to-enterprise using physical cross connects (fiber optic cables) to facilitate private data transfer from one entity to another. This capability is highly valuable because it avoids the security, resiliency, bandwidth, and latency challenges posed by the broader internet, issues that make the broader internet incompatible with the needs of many mission-critical enterprise services, and even where potentially compatible often otherwise unfavorable.

At this juncture of the proceeding, the most salient consideration for the Commission is that within the broad class of large load customers, which itself is extremely diverse, data centers and their business models are highly heterogenous, serve different roles in our economy and national welfare, and have differing needs and capabilities with respect to their interactions with the energy system. We urge the Commission to exercise caution to avoid instituting inadvertent discrimination between data center types and business models as it considers large load interconnection rules. Fortunately, the Commission's long-held prioritization of advancing nondiscriminatory competition to deliver higher performance at least cost, including its efforts to unbundle energy products, will serve the Commission extremely well with the issues presented before it in this docket.

B. Coordination Across Jurisdictions is Key to Both Efficient & Timely Energy System Investments and Large Load Interconnection

We recognize that long-standing jurisdictional issues between the federal, state and local governments are implicated by the Proposed ANOPR. While we do not opine as to appropriate jurisdictions to address energy interconnections of large loads, as an organizing principle we suggest that these issues are too important, too urgent and too interrelated for jurisdictions to act in an uncoordinated fashion.

At a time when our energy system faces its greatest need for investment in decades, efficiently integrating large loads to provide maximum benefit at optimal cost for all grid users is imperative. Study after study has recognized substantially increased investment in our energy system will be required to meet substantially increasing needs.[4] While large loads are not the sole driver of sharply increased load growth forecasts, it is clear the increases in demand they represent will be very significant.[5]

The impact of large load growth is, at the very least, highly likely be transformative to our energy system. If large load growth is recognized as an integral part of our future energy system, rather than as a separate and independent consideration, that transformation could be harnessed to contribute to a more reliable, resilient and affordable energy system. A recent study by Lawrence Livermore National Labs underlines the potential for load growth to dampen, rather than accelerate, rate increases, finding:

> "[G]rowth in recent years (through 2024) has tended to reduce average retail electricity prices. … states with the highest load growth experienced reductions in real prices, whereas states with contracting loads generally saw prices rise."[6]

This empirical result should not be surprising and matches the long understood theoretical relationship between load factor and system costs. Adding new large loads with higher load

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[4] *See, e.g.,* Wilson, Zimmerman & Gramlich, "Strategic Industries Surging: Driving US Power Demand" at 9 – 15 (Grid Strategies, Dec. 2024).

[5] *Id. at 3* (concluding, with respect to load growth, that "[t]he main drivers are investment in data centers and manufacturing").

[6] Wiser, O'Shaughnessy, Barbose, Cappers & Gorman, "Factors Influencing Recent Trends in Retail Electricity Prices in the United States,"38 The Electricity Journal 107516 at 5-6 (Oct. 2025), *available at https://pdf.sciencedirectassets.com/272016/1-s2.0-S1040619025X0004X/1-s2.0-S1040619025000612/main.pdf*

factors than the system as a whole will incrementally increase system load factor and drive down system cost. Load growth from large high load factor loads places strong downward pressure on rates that can offset other factors like inflation, policy, and shifting fuel costs that often push in the opposite direction. Moreover, as energy system investment needs grow separate and apart from large load growth, overall rate pressures would be relieved to the extent large loads can help amortize the cost of those investments.

With the need for energy investments rising and energy affordability concerns rising, if anything, at an even faster pace, it seems obvious the energy system and national interests at issue would be best served if relevant governmental entities worked in concert to develop more uniform approaches to optimally integrating large loads. The pace of integrating large loads needed to serve the national interests identified in the Proposed ANOPR cannot wait for jurisdictional debates to be settled or for lengthy, uncoordinated procedural processes to work their way to conclusion. In light of the stakes at hand, the collective insights and coordinated approaches of all jurisdictions is needed to achieve national, state and local priorities.

Questions:

1. **How Can This Proceeding Maximize Potential Benefits of Large Loads to the Broader Energy System and its Customers?**

A recent trend towards rules and policies seeking to isolate large loads apart from the existing system undermine the basic network effects and returns to scale that have increased system efficiency and reliability and reduced individual customer rates for over a hundred years. At this time, with an energy system under unprecedented stress, it would be particularly perverse to miss opportunities for symbiotic benefit of large load customers and the broader grid.

At a minimum, rather than incentivizing data centers to make investments in energy equipment that focus only on their needs, it seems most prudent to encourage efficient investments that meet large load needs and, where possible, broader electricity customer needs as well. The basic value proposition of interconnected grids is the shared use of infrastructure and services to drive down costs and increase reliability. Stated simply: a generation or storage resource designed solely to serve one specific load behind the meter may miss investment opportunities in resources sufficient to serve that specific load's need as well as the broader grid, at only incremental (or potential no) additional cost. To achieve these important efficiencies, cost allocation for large load interconnections must follow suit: where costs incurred to interconnect a large load serve only to benefit that load, of course the new large load should bear those costs. Where costs incurred provide broader benefit, however, cost allocation should follow the beneficiary pays principle, creating an economic signal that maximizes benefits and minimizes costs for all customers.

With this in mind, one of the most important questions before the Commission is how any interconnection rules it may adopt could promote economically optimal outcomes, while at the same time avoiding complexities and delays that could deter the important national objectives outlined in the Proposed ANOPR and the Secretary's letter to the Commission. We again suggest the best course would be for the Commission to work in concordance with state and local counterparts on interconnection approaches that incentivize optimal economic outcomes, while minimizing the time and complexity of energy interconnections.

2. Flexibility is Critically Important, but Why Adopt Out-of-Market Preferences Solely for Curtailable, Colocated or Hybrid Facilities?

A recent paper, entitled "How DOE's Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility," does an outstanding job of outlining how flexible demand can reduce transmission, distribution and capacity needs, all of which otherwise tend to contribute to the time and cost of energy interconnections for large loads.[7] Without a doubt, demand flexibility has an important role to play in accelerating energy interconnection of large loads at reduced cost, as Principle Seven of the Proposed ANOPR suggests in promoting flexibility by the large load itself or its colocated or hybrid energy resources.[8] However, as the ANOPR also notes, the Commission has advanced unbundling products for many decades.[9]

The purpose of unbundling, of course, is to enable market-driven solutions utilizing competition to spark innovation, reduce costs, and deliver greater efficiencies. It is not clear to us why the flexibility that could clearly offer our energy system such great benefit must, or even should, be either incorporated into the large load itself or necessarily located behind the same interconnection. In fact, artificially constraining flexibility options to the those produced by large load itself could clearly create uneconomic outcomes by as much as *orders of magnitude*, as the Nicholas Institute paper observes.[10]

The universe of flexible resources that large load energy interconnection rules should encompass, and the nature of any incentives granted for inclusion of flexibility within large load energy interconnection requests, are both worthy of serious consideration.

a) The Commission Should Enable Optimal Economic Outcomes, Including All Flexibility Options that Mitigate Interconnection Timelines and Costs

From a grid operations, transmission upgrade, or incremental energy need perspective, it is irrelevant whether a large load is curtailed, backup or other co-located generation reduces the customer's draw from the grid, or other load, generation or storage in sufficient proximity

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[7] Farmer et al, "How DOE's Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility Considerations for Rulemaking" at Table 1 and associated text (Duke Nicholas Institute for Energy, Environment & Sustainability (Nov. 2025)("Nicholas Institute"), available at https://nicholasinstitute.duke.edu/sites/default/files/publications/how-does-proposed-large-load-interconnection-process-could-unlock-benefits-load-flexibility.pdf

[8] Proposed ANOPR at ¶ 24.

[9] *Id. at* ¶ 3.

[10] Nicholson Institute at n. 11, citing Luminary Strategies' finding that "AI data centers' value of lost load could be roughly $250,000/MWh, compared to $20,000/MWh for large industrials, $10,000/MWh for midsized commercial and industrials; and $5,000/MWh for residential customers," and Glen Spry for the proposition that "the value of compute commonly exceeds $750 to $2000/MWh, while compensation for grid services might range from $155 to $340/MWh."

reduces draws from the grid. Enabling the most economic solutions capable of addressing the violations or constraints that would otherwise drive the need for incremental transmission, storage, generation or curtailment would be most consistent with the Commission's long-held economic and market-driven approach. Enabling a wide range of flexible resources to be paired with large load interconnection requests would not only spark competition and innovation; it would enable lower overall costs and potentially broader benefits.

For example, while some data centers may be located in areas that are amenable to on-site generation and/or storage, others, such as those that must be located in very close proximity to financial sector, national security, or other critical uses that demand the absolute minimum security risk and data latency, may not. Those data centers located in dense metropolitan areas should be able to tap into the wealth of flexibility opportunities that surround them, creating a mutual benefit for those commercial, industrial or even residential users that could receive enhanced energy equipment or could easily absorb curtailment at a much lower economic and performance cost.

Failure to enable interconnection customers to avail themselves of the least expensive effective options would not only be contrary to the Commission's economic precepts: it would inadvertently introduce discrimination between large load interconnection customers. Penalizing those large loads that, for national security, financial sector or other reasons must maintain extremely high up-time, and for those same reasons are geographically constrained to areas that have limited ability to incorporate backup or co-located generation, would be contrary to national interests.

To enable the most economic solutions to issues that might otherwise require timely and expensive incremental energy system investments, we suggest that large load interconnection customers should be allowed to include what could be termed "aggregated resource" proposals in their interconnection requests. So long as a large load is contracted with and can deliver a solution that resolves or mitigates the constraints and violations that would otherwise trigger incremental energy system costs, that aggregated proposal should be studied as part of the large load interconnection request. This approach would necessitate study transparency requirements that allow large load interconnection customers to understand the specific constraints and resulting incremental energy system investments their energy interconnection might cause. Interconnection processes should give large load interconnection customers an opportunity to respond to study results and bring forth solutions that could avoid, defer, or mitigate the need to build upgrades or make other incremental investments.

b) Flexibility is Its Own Reward: Advancing Timelines and Reducing Costs Are Appropriate Market-Driven Incentives, While Processing Priorities Are Not.

Flexibility, including demand response, co-located or hybrid generation and storage, or other measures incorporated "aggregated resource" packages, all have much to offer in reducing costs and accelerating interconnection timelines, and should be encouraged. Providing that study frameworks accurately model and value the relevant operational characteristics of the proposed large load along with its aggregated flexibility resources, it is not clear to us why study queue prioritization or any other exogenous benefit is necessary or appropriate.

The incentives of reduced interconnection time and expense resulting from deferred, avoided or minimized system upgrades are powerful incentives that should drive technological and commercial innovation. Ultimately, flexibility, hybrid pairings, or any other aggregated interconnection proposal can and should provide their own reward. By collectively presenting a lower burden (or even a valuable asset) to the energy system, the interconnection customer should reduce its costs and benefit from faster interconnection, without need for an out-of-market administrative processing priority. Non-market advantages of one type of flexibility over another, in contrast, bears great risk of unintended discrimination.

We recommend against introducing study prioritization frameworks. In our experience, operating in 37 countries around the world—including many markets where such frameworks are in place—such approaches introduce considerable room for fluctuating policy objectives to muddy economic signals, and to increase uncertainty, associated financing costs, and interconnection timing. Consistent with the Commission's precedent, load and supply interconnection queue processes are best grounded in specific technical and financial criteria that enable competition and drive innovation, efficiencies and reduced cost.

V. Conclusion

Equinix again applauds the Department of Energy and the Commission for their initiative in launching this proceeding, and appreciates this opportunity to address such urgent and important issues for our energy system and our economy. We ask that the Commission keeps the following considerations in mind as it weighs potential large load interconnection processes.

- Given the transformative impact large loads are certain to have on our energy system, as well as the rapidly growing needs for energy system investments, large load interconnection processes should enable system optimal investments consistent with minimizing interconnection timelines and costs.
- Given the urgency of these issues, the Commission and all relevant jurisdictions should strive to work together, adopting more uniform approaches that maximize benefits.
- Given the great value of increasing flexibility, the heterogenous nature of large loads and the risk of introducing inadvertently discriminatory measures, as well as the benefits of enabling a broad array of customers and resources to contribute flexibility, large load interconnection processes should allow interconnection customers to offer aggregate resource packages that hold reasonable potential to minimize interconnection costs and timelines.
- Given that flexibility is its own reward, by avoiding or minimizing costs and time of building incremental upgrades, no administrative study prioritization or other exogenous incentives are warranted, and risk inadvertent discrimination.

We thank the Commission for its consideration of these comments.

Respectfully submitted,

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